

Local Subsidies Fraud Risk - Model Comparison

Note: Korean narrative is in docs/metrics_guide.md. This PDF summarizes aggregate metrics only (no raw rows).

Train period: 202401-202506

Test period: 202507-202512

Risk score range: 0-1000 (higher = higher fraud risk).

Reports contain aggregate statistics only. No raw PII rows.

See docs/metrics_guide.md for Korean explanations of metrics.

Per-algorithm reports: outputs/reports/{algorithm}/

Model metrics (aggregate)

algorithm=catboost_v1, Accuracy=0.9880541750552592, Precision=0.21994408201304752, Recall=0.9535353535353536, F1=0.35744036349867475, ROC_AUC=0.9892893031100048

algorithm=stacked_ensemble_v1, Accuracy=0.9549620577510595, Precision=0.07019076743847386, Recall=0.9737373737373738, F1=0.13094267861994024, ROC_AUC=0.9904099948504206

algorithm=easy_ensemble_v1, Accuracy=0.7351293133790424, Precision=0.010080219548237282, Recall=0.7717171717171717, F1=0.01990049751243781, ROC_AUC=0.8431200100294425

algorithm=gradient_boosting_v1, Accuracy=0.9703149417843416, Precision=0.09277899343544858, Recall=0.8565656565656565, F1=0.16742349457058242, ROC_AUC=0.9626496166142824

algorithm=random_forest_v1, Accuracy=0.9994931647636881, Precision=0.904397705544933, Recall=0.9555555555555556, F1=0.9292730844793713, ROC_AUC=0.9916065924034108